Tugas Machine Learning

Muhammad Zharfan Bimantoro

2602624891

Tugas

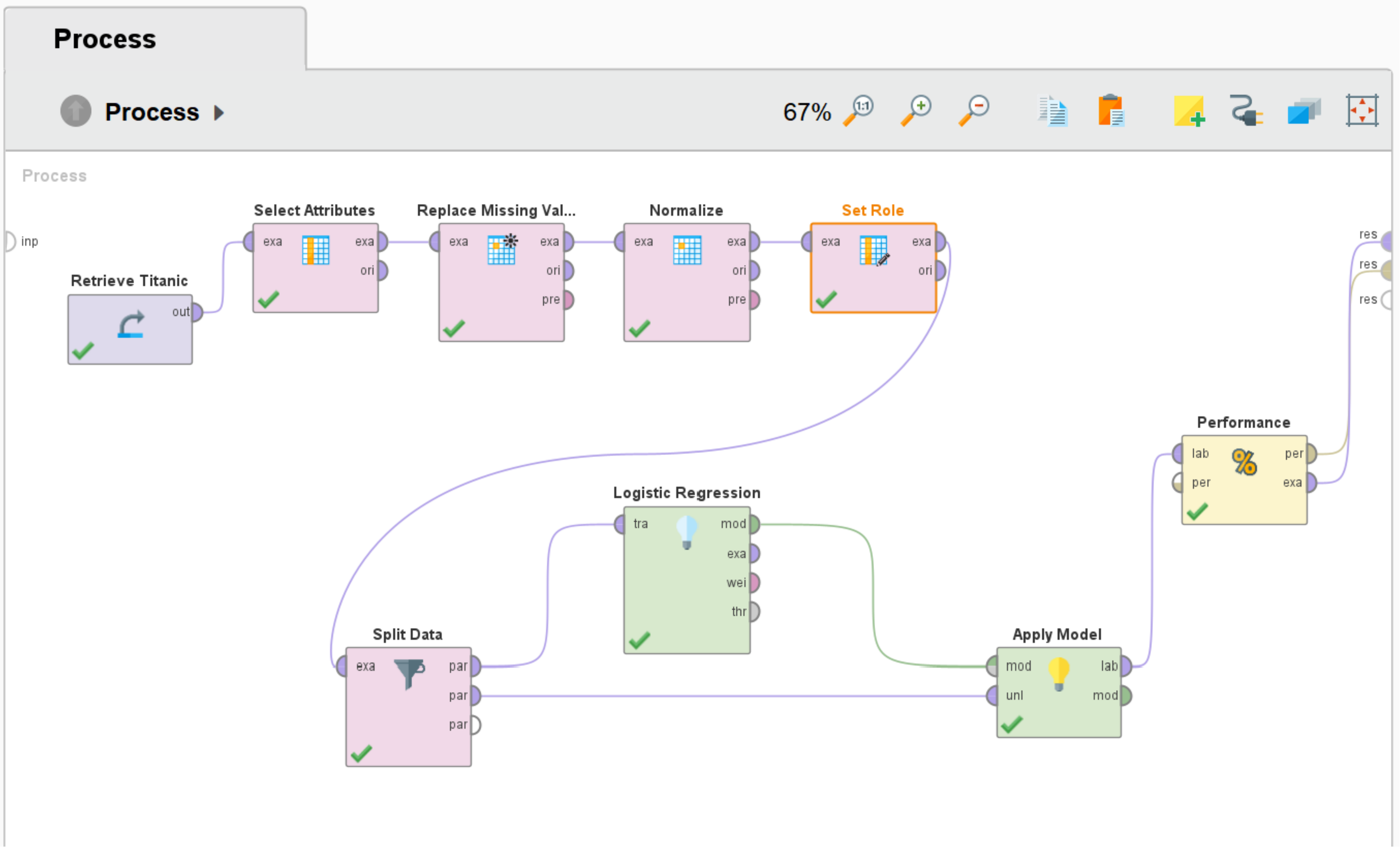
1. Klasifikasi dengan 1 metode untuk titanic dengan menggunakan Rapid Miner dan Python. Bandingkan hasilnya.Tunjukkan performancenya

2. Klasifikasi dengan 1 metode untuk iris dengan menggunakan Rapid Miner dan Python. Bandingkan hasilnya. Tunjukkan performancenya. Pahami multi class dan binary class

3. https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic. Lakukan feature selection dengan PCA dengan 1 metode klasifikasi. Rapid Miner dan program

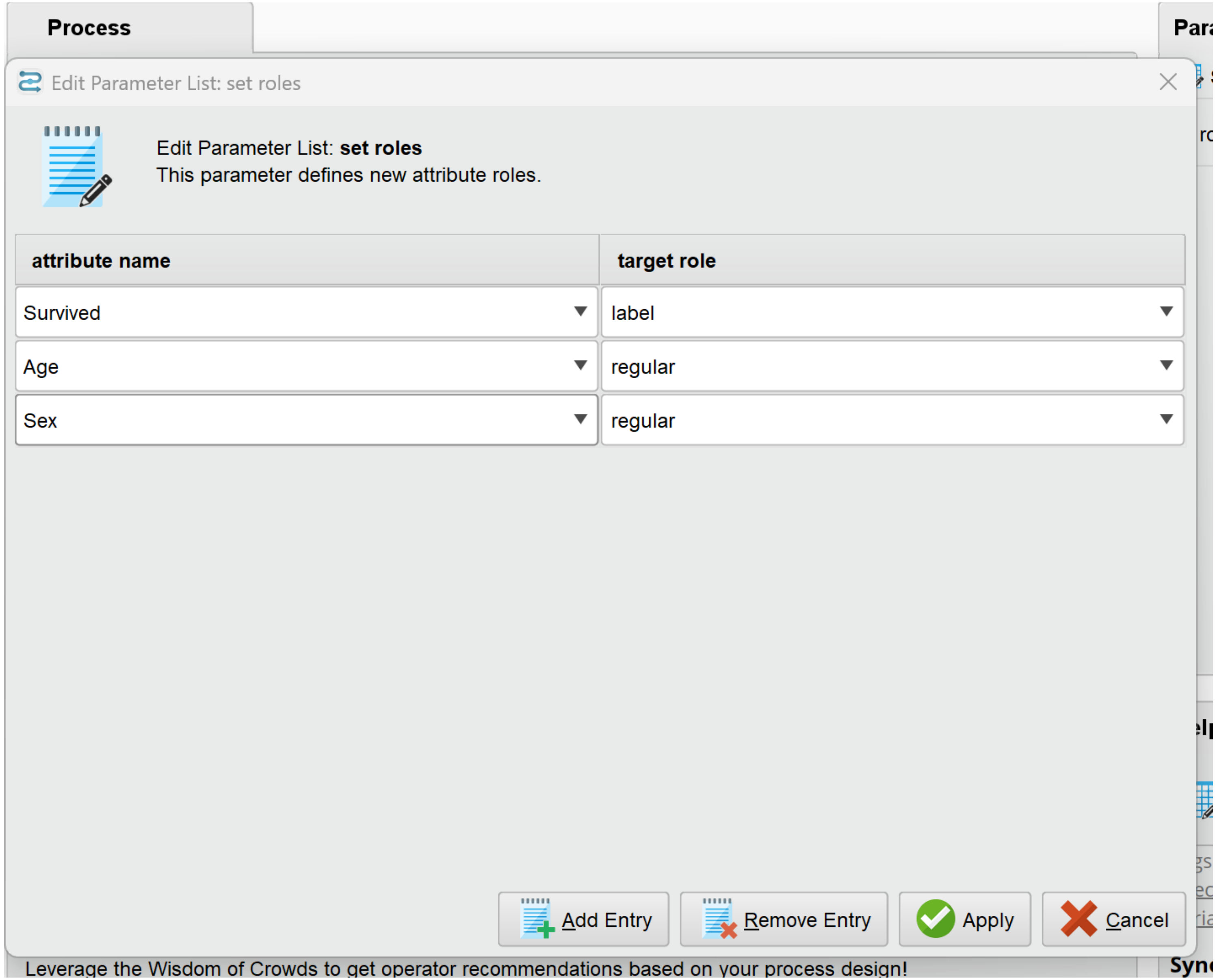
Titanic Dataset

- Design



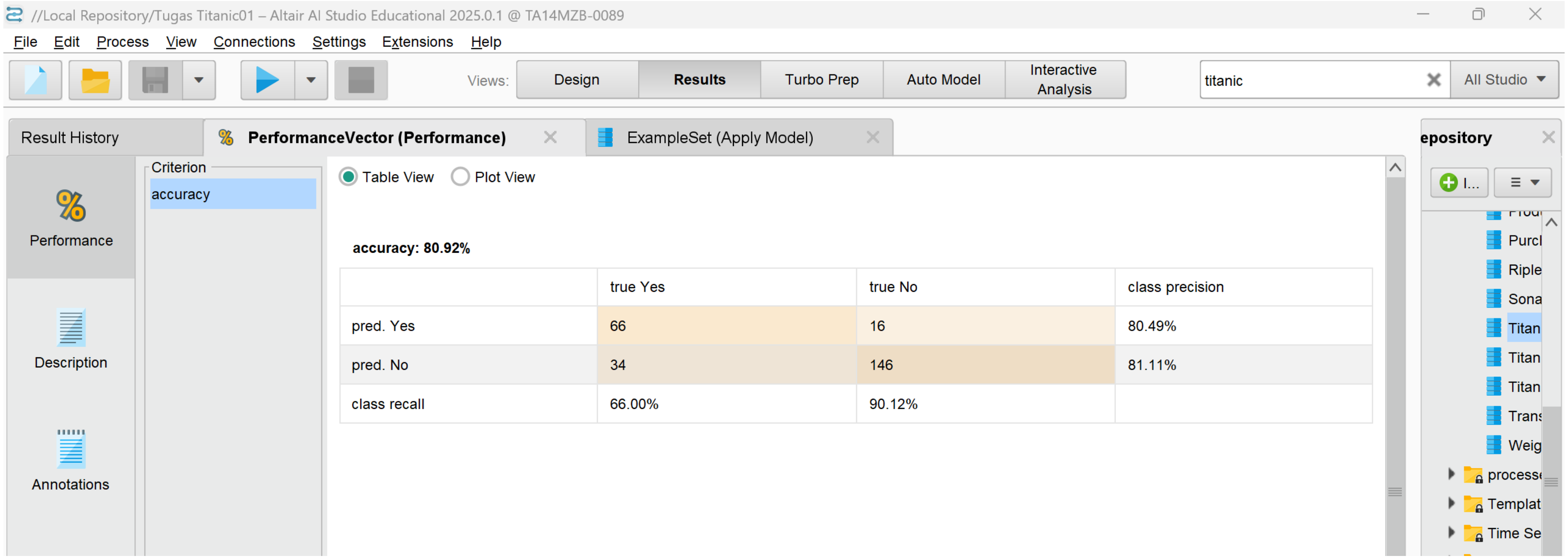
Titanic Dataset

- Design



Titanic Dataset

- Result



	true Yes	true No	class precision
pred. Yes	66	16	80.49%
pred. No	34	146	81.11%
class recall	66.00%	90.12%	

Titanic Dataset

- Perbandingan dengan Python

accuracy: 80.92%

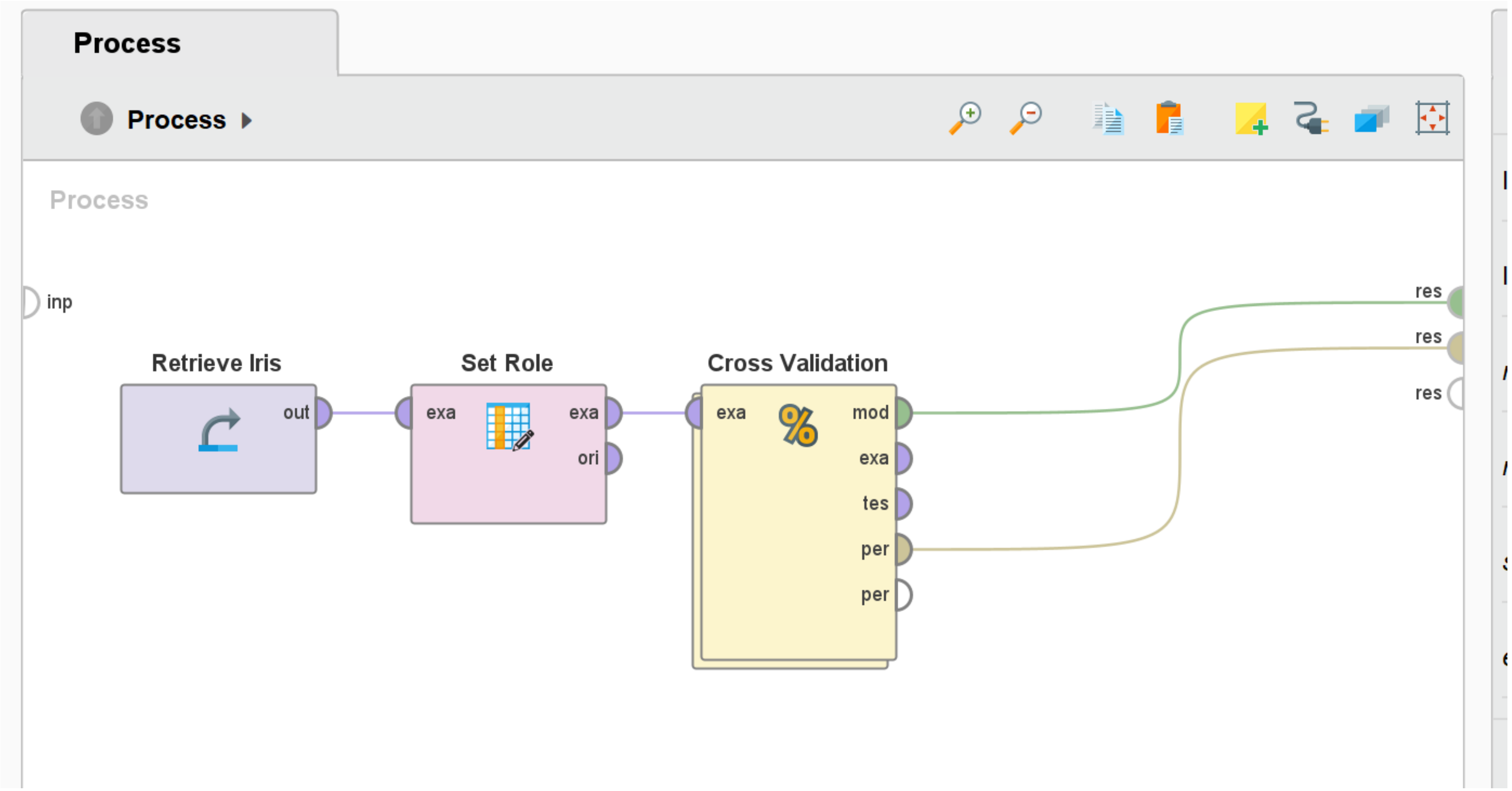
	true Yes	true No	class precision
pred. Yes	66	16	80.49%
pred. No	34	146	81.11%
class recall	66.00%	90.12%	

```
Accuracy: 0.9580

Classification Report:
              precision    recall  f1-score   support

           0       0.93      1.00      0.96       144
           1       1.00      0.91      0.95       118

    accuracy                           0.96       262
   macro avg       0.96      0.95      0.96       262
weighted avg       0.96      0.96      0.96       262
```

Hasil yang diraih ketika menggunakan python rata-rata lebih tinggi

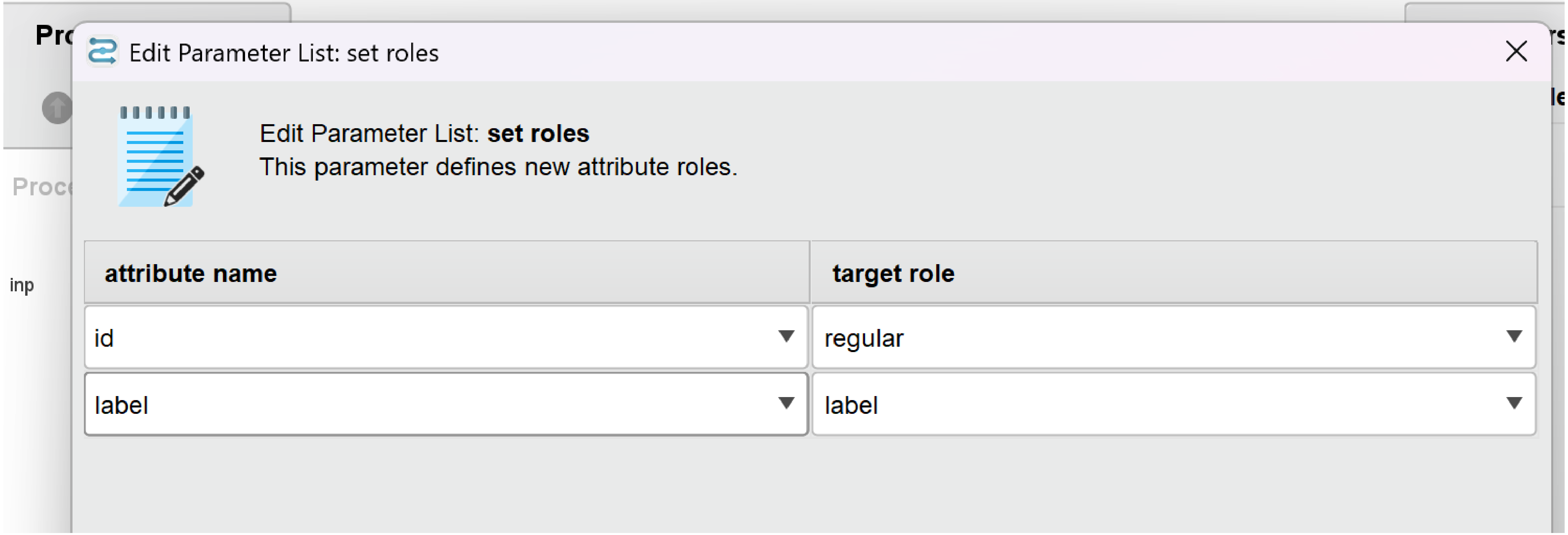
Iris Dataset

- Design



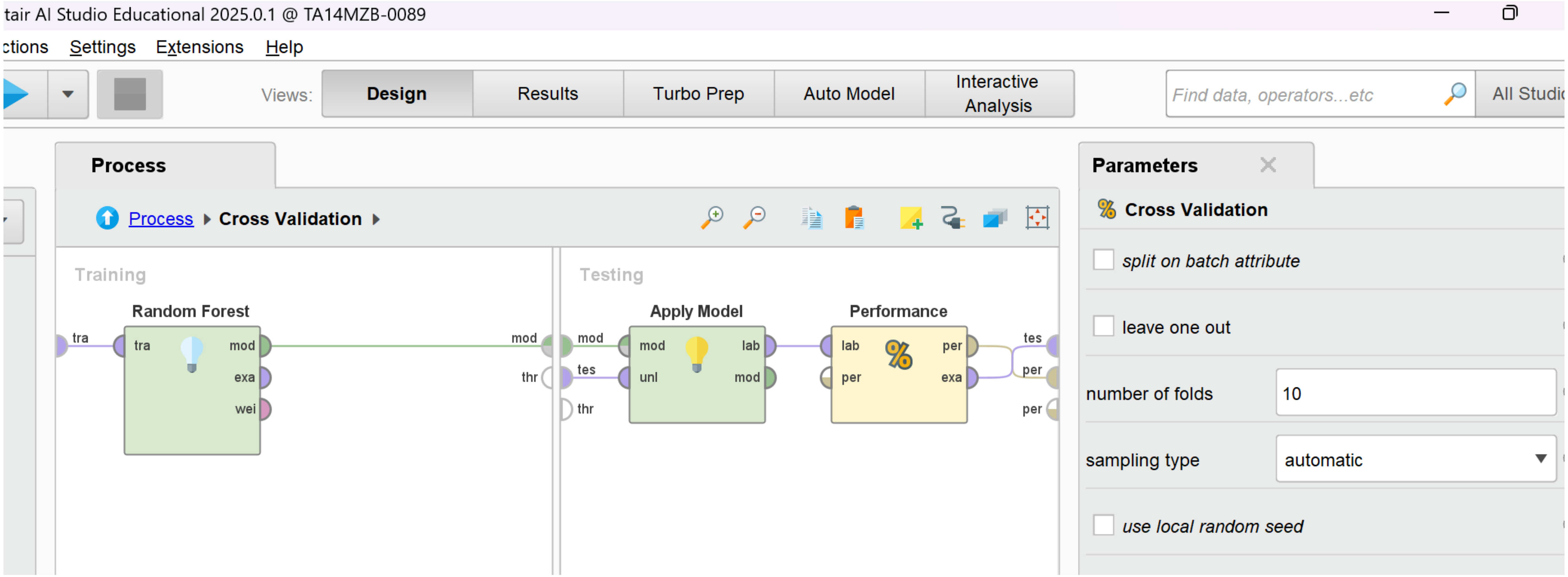
Iris Dataset

- Design



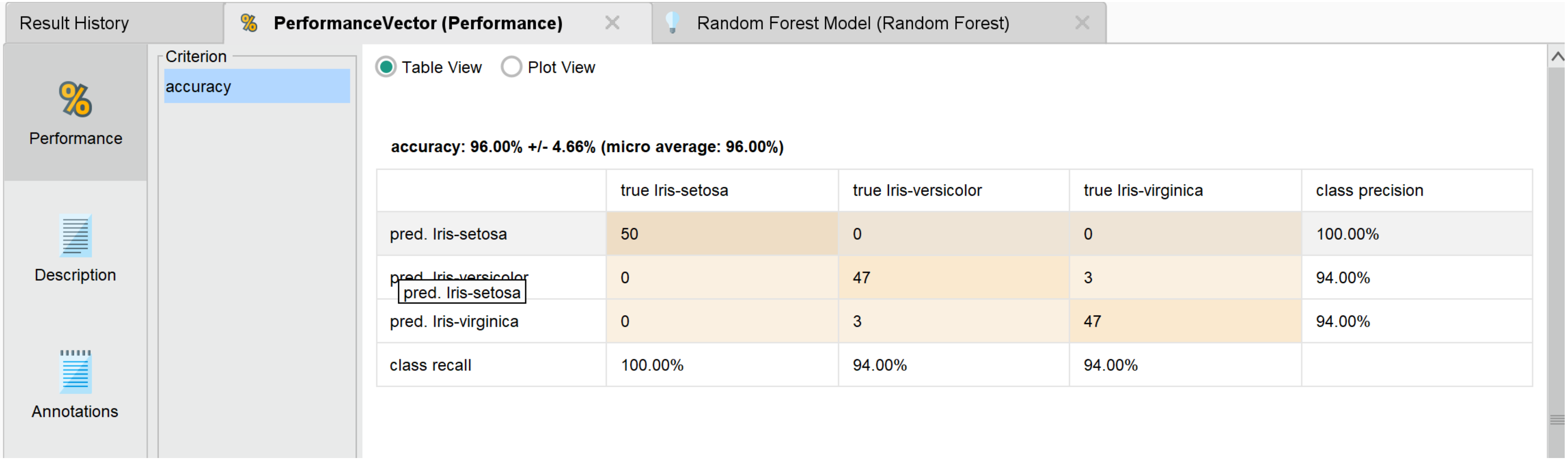
Iris Dataset

- Design



Iris Dataset

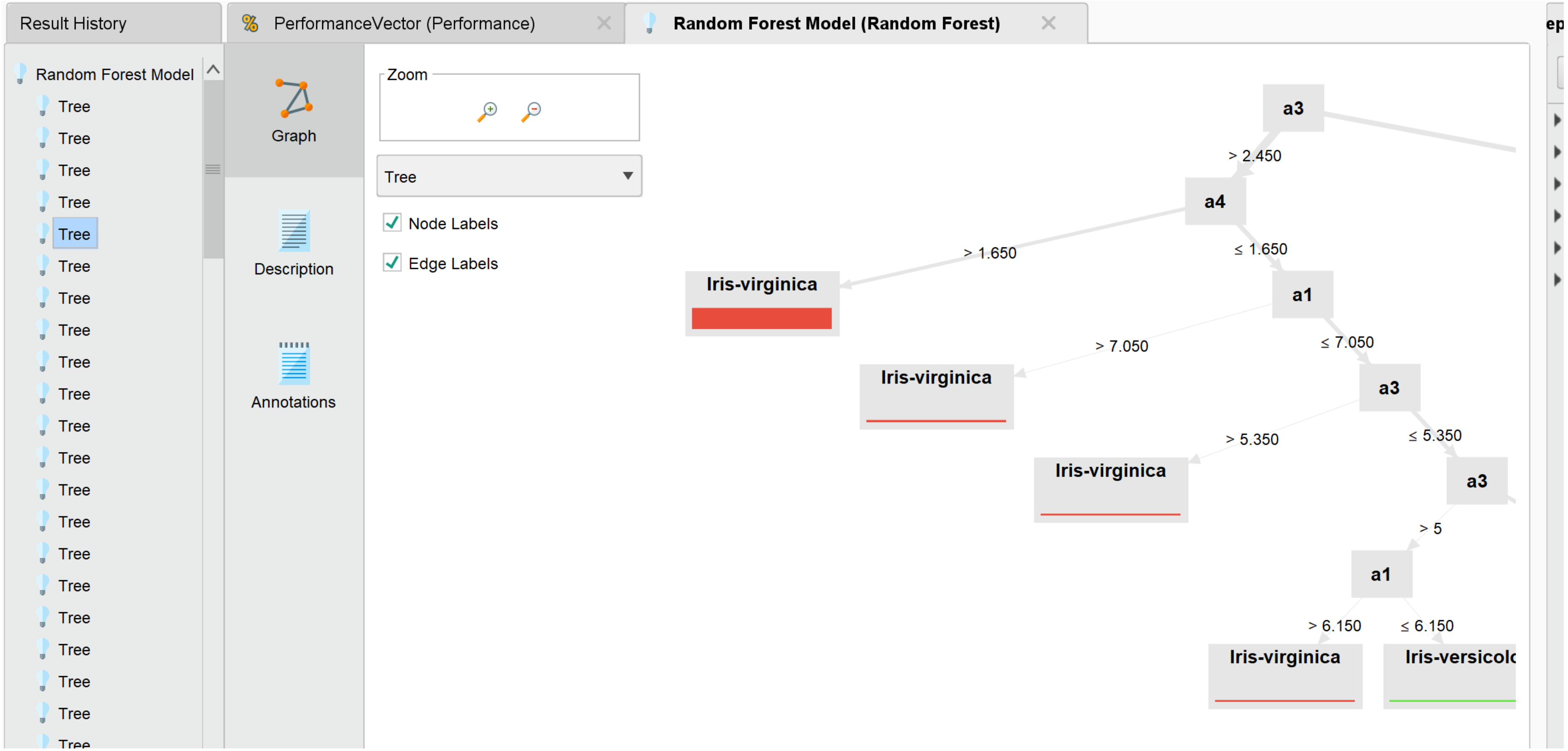
- Result



	true Iris-setosa	true Iris-versicolor	true Iris-virginica	class precision
pred. Iris-setosa	50	0	0	100.00%
pred. Iris-versicolor	0	47	3	94.00%
pred. Iris-virginica	0	3	47	94.00%
class recall	100.00%	94.00%	94.00%	

Iris Dataset

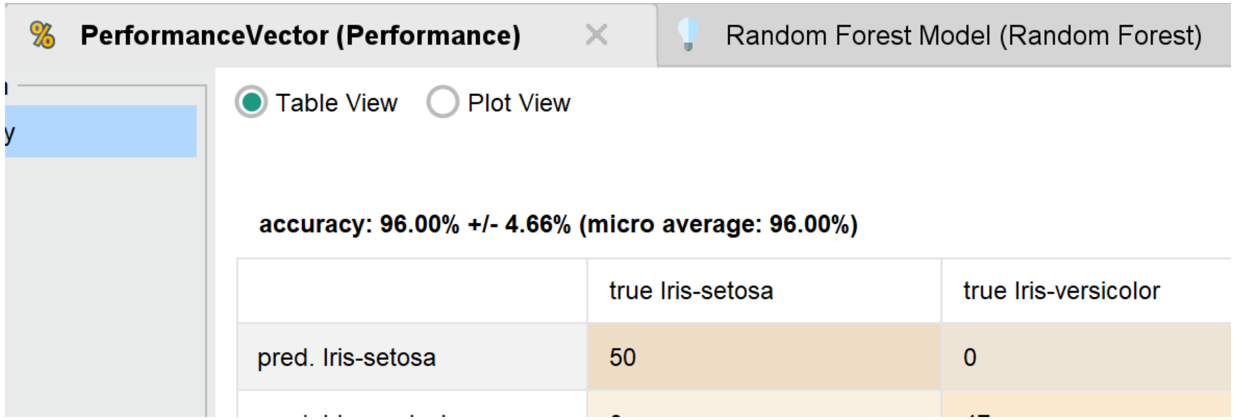
- Result



Iris Dataset

- Perbandingan dengan Python

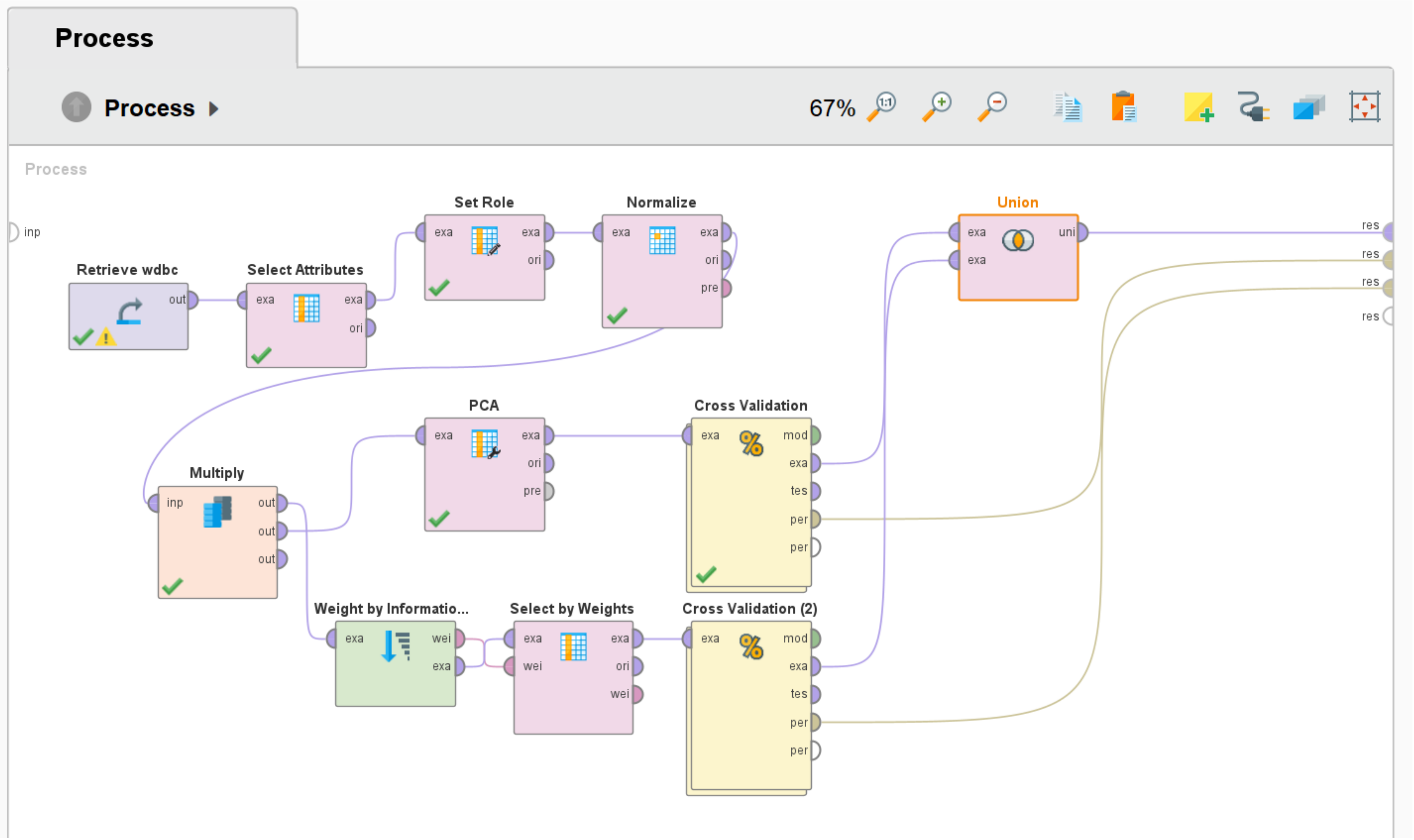
```
Model: RandomForestClassifier
Accuracy: 0.9556

Classification Report:
                 precision    recall  f1-score   support

    Iris-setosa       1.00      1.00      1.00        16
Iris-versicolor       0.91      0.91      0.91        11
 Iris-virginica       0.94      0.94      0.94        18

       accuracy                           0.96        45
      macro avg       0.95      0.95      0.95        45
   weighted avg       0.96      0.96      0.96        45

Model: SVC
Accuracy: 0.9778

Classification Report:
                 precision    recall  f1-score   support

    Iris-setosa       1.00      1.00      1.00        16
Iris-versicolor       0.92      1.00      0.96        11
 Iris-virginica       1.00      0.94      0.97        18

       accuracy                           0.98        45
      macro avg       0.97      0.98      0.98        45
   weighted avg       0.98      0.98      0.98        45
```



Ketika menggunakan rapidminer, didapatkan hasil akurasi 96%, sementara Ketika menggunakan Python, menggunakan model yang sama yaitu random forest didapatkan hasil yang lebih rendah, tetapi didapatkan hasil akurasi 97% Ketika menggunakan model SVC / Support Vector Machine Classifier

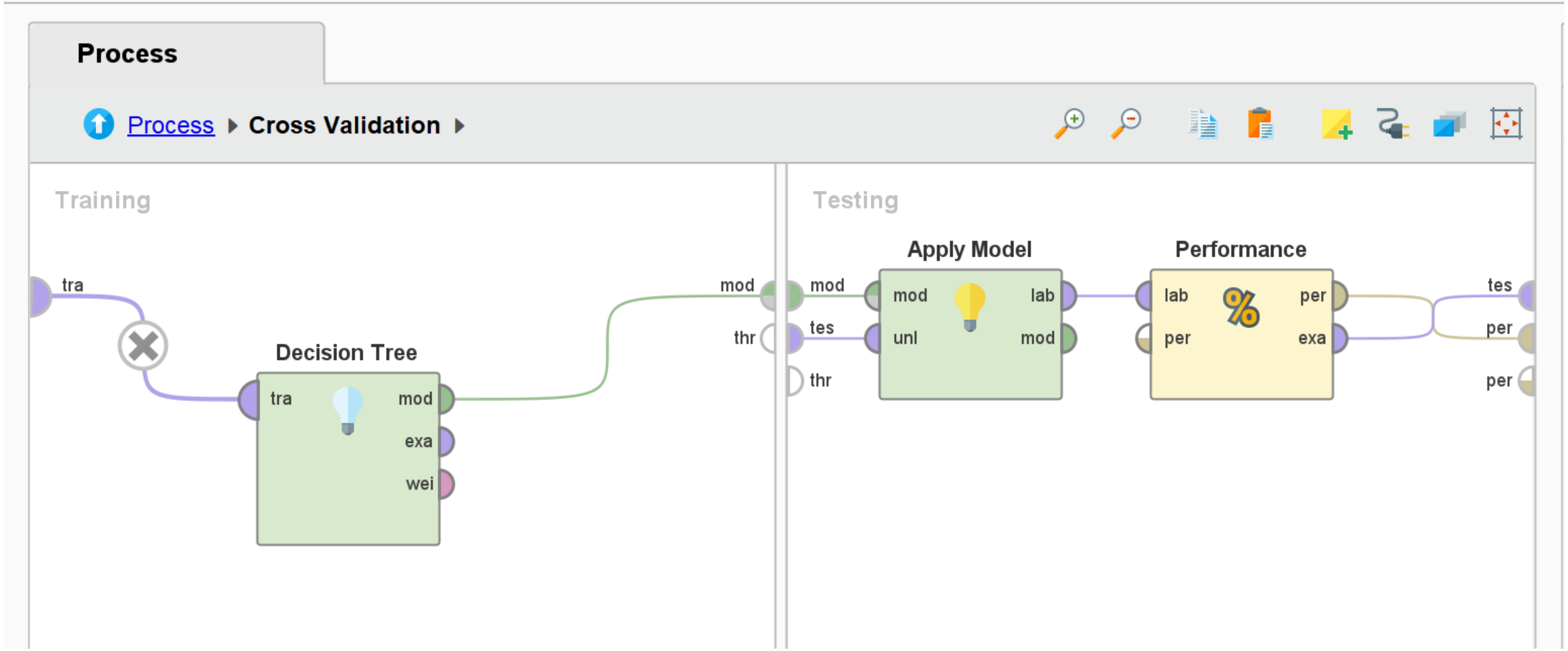
Wisconsin Breast Cancer Dataset

- Design



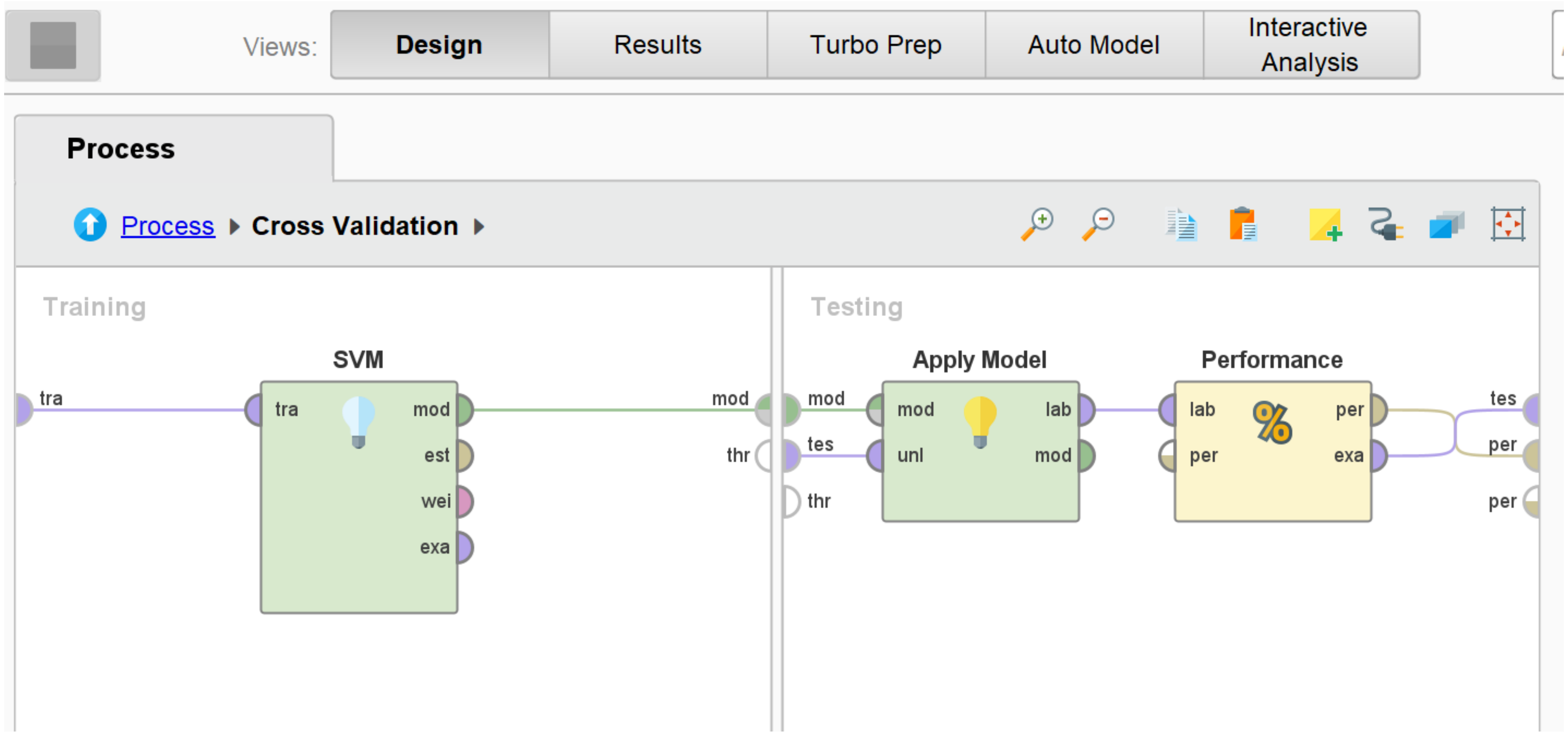
Wisconsin Breast Cancer Dataset

- Design



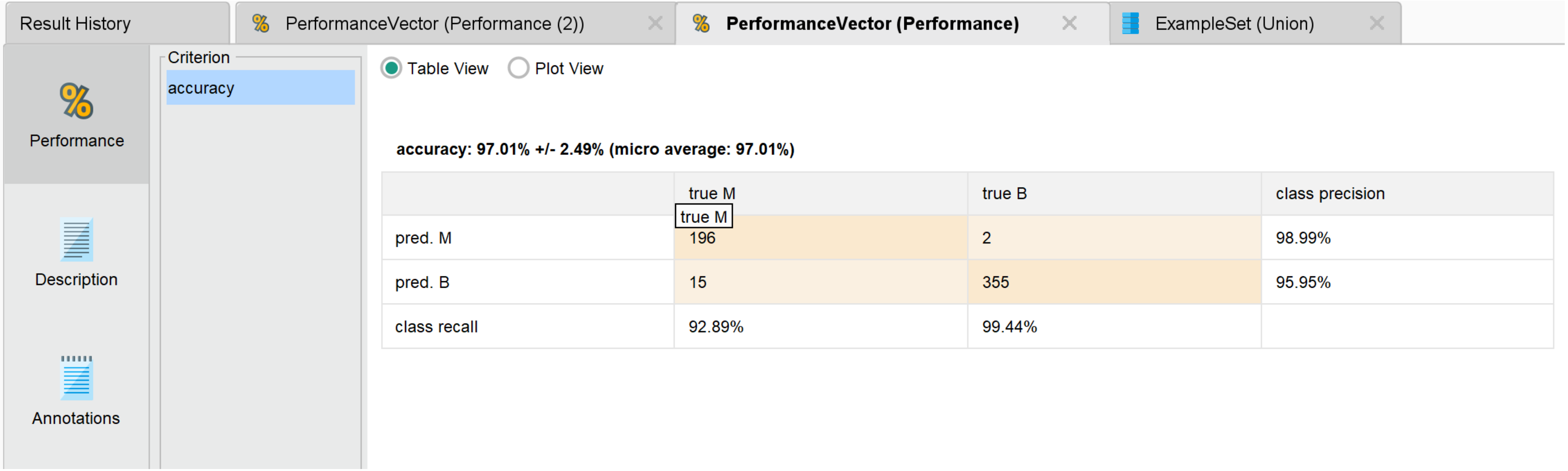
Wisconsin Breast Cancer Dataset

- Design



Wisconsin Breast Cancer Dataset

- Result (Principal Component Analysis)

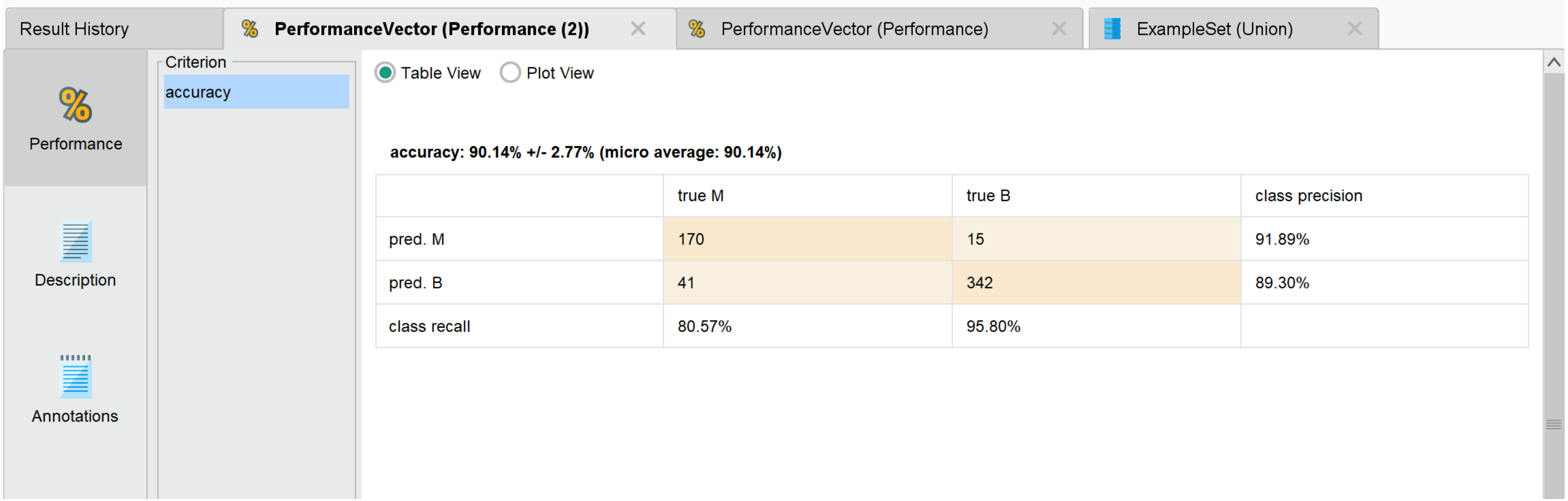


accuracy: 97.01% +/- 2.49% (micro average: 97.01%)

	true M	true B	class precision
pred. M	196	2	98.99%
pred. B	15	355	95.95%
class recall	92.89%	99.44%	

Wisconsin Breast Cancer Dataset

- Result (Select by Weight)

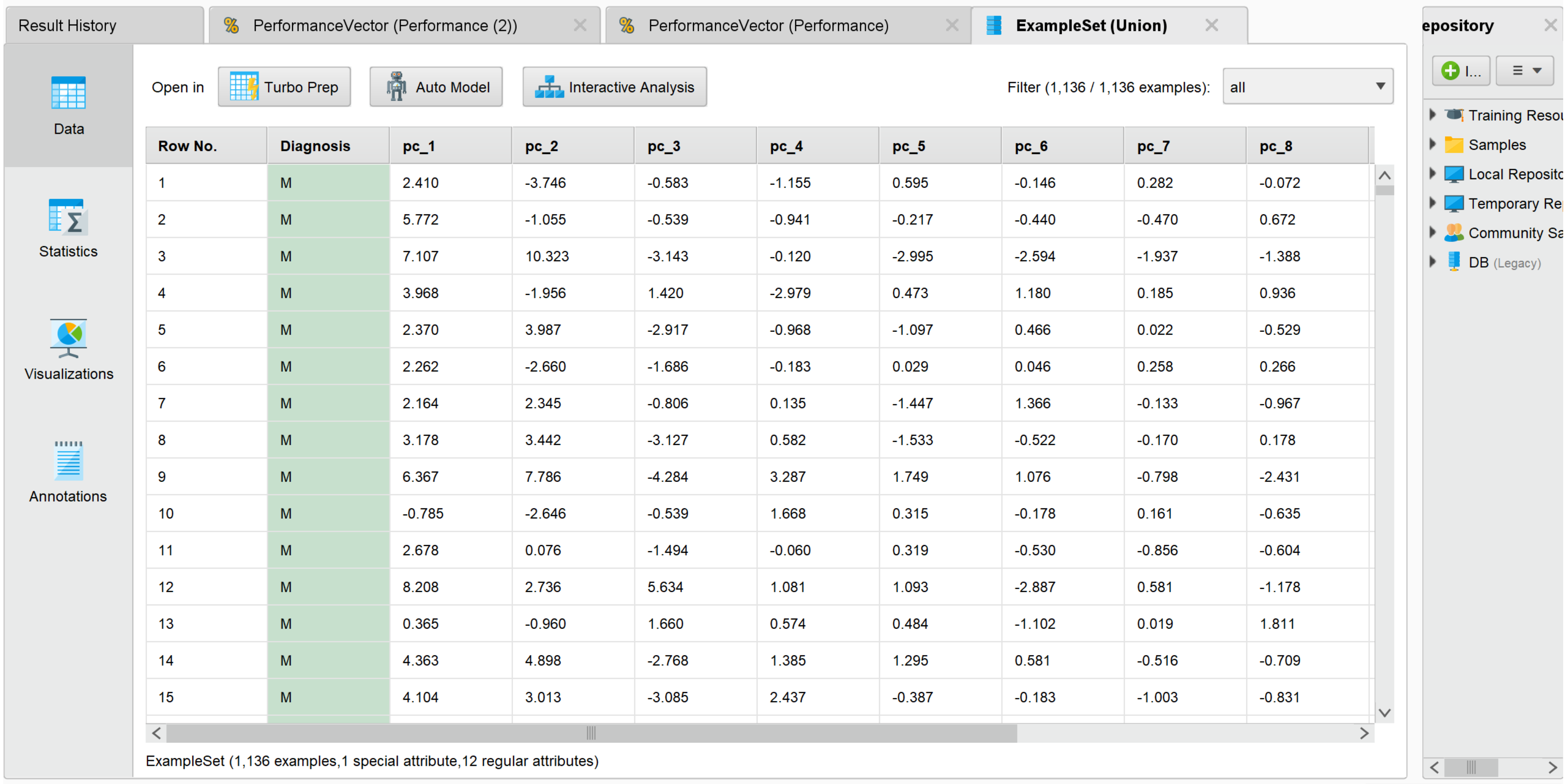


accuracy: 90.14% +/- 2.77% (micro average: 90.14%)

	true M	true B	class precision
pred. M	170	15	91.89%
pred. B	41	342	89.30%
class recall	80.57%	95.80%	

Wisconsin Breast Cancer Dataset

- Result



Row No.	Diagnosis	pc_1	pc_2	pc_3	pc_4	pc_5	pc_6	pc_7	pc_8
1	M	2.410	-3.746	-0.583	-1.155	0.595	-0.146	0.282	-0.072
2	M	5.772	-1.055	-0.539	-0.941	-0.217	-0.440	-0.470	0.672
3	M	7.107	10.323	-3.143	-0.120	-2.995	-2.594	-1.937	-1.388
4	M	3.968	-1.956	1.420	-2.979	0.473	1.180	0.185	0.936
5	M	2.370	3.987	-2.917	-0.968	-1.097	0.466	0.022	-0.529
6	M	2.262	-2.660	-1.686	-0.183	0.029	0.046	0.258	0.266
7	M	2.164	2.345	-0.806	0.135	-1.447	1.366	-0.133	-0.967
8	M	3.178	3.442	-3.127	0.582	-1.533	-0.522	-0.170	0.178
9	M	6.367	7.786	-4.284	3.287	1.749	1.076	-0.798	-2.431
10	M	-0.785	-2.646	-0.539	1.668	0.315	-0.178	0.161	-0.635
11	M	2.678	0.076	-1.494	-0.060	0.319	-0.530	-0.856	-0.604
12	M	8.208	2.736	5.634	1.081	1.093	-2.887	0.581	-1.178
13	M	0.365	-0.960	1.660	0.574	0.484	-1.102	0.019	1.811
14	M	4.363	4.898	-2.768	1.385	1.295	0.581	-0.516	-0.709
15	M	4.104	3.013	-3.085	2.437	-0.387	-0.183	-1.003	-0.831

Wisconsin Breast Cancer Dataset

- Perbandingan dengan Python

```
SVM Accuracy: 0.9766
Classification Report:
              precision    recall  f1-score   support

           B       0.97      0.99      0.98       103
           M       0.98      0.96      0.97        68

    accuracy                           0.98       171
   macro avg       0.98      0.97      0.98       171
weighted avg       0.98      0.98      0.98       171
```

PerformanceVector (Performance (2)) | PerformanceVector (Performance)

Table View | Plot View

accuracy: 90.14% +/- 2.77% (micro average: 90.14%)

	true M	true B
pred. M	170	15
pred. B	41	342
class recall	80.57%	95.80

ceVector (Performance (2)) | PerformanceVector (Performance) | ExampleSet (Union)

Table View | Plot View

accuracy: 97.01% +/- 2.49% (micro average: 97.01%)

	true M	true B	class precision
pred. M	196	2	98.99%
pred. B	15	355	95.95%
class recall	92.89%	99.44%	

Akurasi SVM menggunakan Python dan Rapidminer relatif sama, tetapi terdapat penurunan performa Ketika menggunakan feature selection menjadi 90% akurasi.